## Exception handling

```
a = 5
b = 0
print(a/b)
```

zero division error

```
try:
    # risky code
except:
    # handle error.
```

```
try:
    print(5/0)
except:
    print("error occured")
```

| Error | Meaning |
|---|---|
| → Zero division error | divide by zero |
| → Valueerror | wrong value type |
| → Typeerror | wrong datatype |
| → Indexerror | index out of range |
| → Keyerror | dict key missing |
| → filenotfounderror | file not found |

Ex:-

```
x = int("hello")
```

```
try:
    x = int("hello")
except:
    print("Invalid conversion").
```

```
try:
    ___
except:
    ___
finally:
    ___
```

```
try:
    print(10/0)
except:
    print("error occured")
finally:
    print("This always runs")
```

```
try:
    exist:
    else:
```

raise exception("message")

## Assert Statement

↳ test assumptions.

assert condition, "message"

## Regex → Regular expression

↓ Pattern

↓ sequence of characters that defines a search pattern.

→ searching
→ matching
→ extracting
→ replacing text

| Field | Use |
|---|---|
| NLP | text cleaning |
| DS | preprocessing |
| Web scrapping | extracting data |
| fraud detection | pattern matching |
| chatbots | input detection |

→ Built in module (re)

```
import re
```

1. re.search() — find firstmatch
2. re.match() — match from start
3. re.findall() — get all matches
4. re.sub() — replace
5. re.split() — split

## Meta characters (♡)

| Symbol | Meaning |
|---|---|
| . | any character |
| ^ | start |
| $ | end |
| * | 0 or more |
| + | 1 or more |
| ? | optional |
| [] | set |
| {} | exact count |
| \ | escape |

| Pattern | Meaning |
|---|---|
| \d | digits |
| \D | not digits |
| \w | word |
| \W | not word |
| \s | space |
| \S | no space. |

## String method

1. text = "Hello world"
   print(text.lower()) → hello world
   print(text.upper()) → HELLO WORLD
2. text = "  hello  "
   print(text.strip()) → hello
3. text = "I love Java"
   print(text.replace("Java", "python"))
   ↳ I love Python
4. s = "apple, banana, orange"
   print(s.split(", "))
   ['apple', 'banana', 'orange']
5. words = ['I', 'love', 'Python']
   print(" ".join(words))
   I love python.